

What published machine-learning models on public actigraphy cohorts claim, and what honest evaluation shows: a PROBAST+AI / TRIPOD+AI appraisal and empirical re-evaluation on DEPRESJON, PSYKOSE and HYPERAKTIV

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Code, run manifests and generated tables: <https://github.com/rogerpanel/cv/tree/main/mlmh-appraisal>

Abstract

Background. Three open actigraphy cohorts from the same Norwegian group, DEPRESJON (depression), PSYKOSE (schizophrenia) and HYPERAKTIV (ADHD), now harmonised as OBF-Psychiatric, have become standard benchmarks for machine-learning (ML) detection of mental disorders, with reported accuracies rising from the dataset authors' 0.7 to above 0.9. Whether those gains reflect better models or weaker evaluation has not been examined. **Methods.** Part I is a systematic review of every study that develops or validates a supervised model on these cohorts (citation chasing of the four dataset papers plus database searches; two reviewers; PROBAST+AI risk of bias and TRIPOD+AI reporting), with pre-specified outcomes: unit of splitting for repeated measures, external validation, calibration reporting, AUROC versus accuracy-only reporting, subject-level events per predictor, double counting of the control group shared by DEPRESJON and PSYKOSE, and code availability. Part II re-evaluates the same cohorts with fixed models (logistic regression, random forest, XGBoost, 1D-CNN), ten seeds and subject-level bootstrap intervals, under (E1) record-wise versus subject-wise cross-validation, (E2) frozen cross-cohort transfer with the shared controls assigned to one cohort, (E3) calibration for every model, (E4) sex strata and (E5) transdiagnostic and five-class tasks on OBF-Psychiatric. **Results.** Part I: [TODO: n studies; proportions]. Part II: record-wise splitting inflated AUROC by 0.065 to 0.114 (DEPRESJON), 0.020 to 0.044 (PSYKOSE) and 0.130 to 0.169 (HYPERAKTIV, honest estimate 0.44 to 0.45, leaky 0.60 to 0.65 at subject level); PSYKOSE-trained models lost 0.13 to 0.21 AUROC on DEPRESJON with calibration slopes 0.27 to 0.63; honest slopes were 0.53 to 0.95 and no published model reports them; subject-level events per predictor were 0.6 to 0.8; the 32 PSYKOSE controls are byte-identical to the DEPRESJON controls. **Conclusions.** Reported progress on these benchmarks is largely an artefact of evaluation design. We provide a subject-wise, calibrated, cross-cohort baseline with public code against which future claims can be checked.

1 Introduction

Public benchmark datasets accelerate a field and, when their evaluation conventions are loose, inflate it. The DEPRESJON [1], PSYKOSE [2] and HYPERAKTIV [3] actigraphy datasets, released between 2018 and 2021 and harmonised in 2025 as OBF-Psychiatric [4], are the most widely used open data for ML detection of mental disorders from wearable activity. The dataset authors reported a leave-one-patient-out F1 of 0.73 for depression [1]; the literature that followed reports accuracies of 0.85 to 0.96 for depression [5, 6] and about 0.94 for schizophrenia [7], usually

with day- or window-level cross-validation, never with calibration, and never with validation on a second cohort. A field-wide appraisal of ML models in psychiatry found calibration reported in 22% of models and adequate internal validation in 11% [13]; PROBAST+AI [14] and TRIPOD+AI [15] now specify what an ML prediction study must do and report, including explicit attention to data leakage.

This paper pairs two things that are usually published apart. Part I is a closed-scope systematic review: every study that develops or validates a model on these cohorts, appraised with PROBAST+AI and TRIPOD+AI, with outcomes chosen because they can be checked against the data (unit of splitting, external validation, calibration, sample size, and whether the control group shared between DEPRESJON and PSYKOSE was double counted). Part II re-evaluates the cohorts under the designs the review scores as high or low risk, with identical models and seeds, so that each appraisal outcome has a measured cost. A closed scope makes the review complete rather than sampled, and puts reviewer and re-analyst on the same data; the field-wide appraisal across all modalities is reported separately [17].

2 Part I: systematic appraisal of the published models

2.1 Eligibility

Any peer-reviewed article or indexed conference paper (2018 to [TODO: date], English) that trains or evaluates a supervised model to predict participant group, symptom score or state from DEPRESJON, PSYKOSE, HYPERAKTIV or OBF-Psychiatric activity data, reporting at least one performance metric. Excluded: reviews, dataset descriptors without a model, studies using the cohorts only for unsupervised analysis or feature description.

2.2 Search

Forward citation chasing of the four dataset papers in Scopus, Web of Science and Google Scholar, plus database searches for the dataset names (“Depresjon” OR “Psykose” OR “Hyperaktiv” OR “OBF-Psychiatric”) in PubMed, Scopus, Web of Science, IEEE Xplore and the ACM Digital Library, run on [TODO: date]; counts in Supplementary Table S3. Two reviewers screened and extracted independently (Rayyan; Cohen’s κ); disagreements resolved by discussion. Preprints were included and analysed as a stratum. [TODO: PROSPERO / OSF registration.]

2.3 Appraisal and outcomes

PROBAST+AI (four domains) and TRIPOD+AI (27 items) in duplicate. Pre-specified outcomes per study: (1) unit of splitting (participant, day/window, unstated); (2) any external validation; (3) calibration reported; (4) AUROC reported or accuracy only; (5) subject-level events per candidate predictor; (6) whether both DEPRESJON and PSYKOSE were used and, if so, whether the shared control group was recognised; (7) code availability; (8) best reported accuracy/AUROC, so that the distribution of claims can be plotted against the honest re-evaluation of Part II.

2.4 Results of Part I

[TODO: PRISMA flow (Supplementary Figure S1). Study characteristics table. PROBAST+AI domain judgements (stacked bars). TRIPOD+AI adherence per item. Outcome proportions with Wilson 95% CIs. Figure: reported accuracy/AUROC per study, coloured by unit of splitting, with the Part II subject-wise and record-wise estimates drawn as horizontal bands.] Seed studies

identified before the formal search (not a substitute for it) are listed in Supplementary Table S4: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12].

3 Part II: empirical re-evaluation

3.1 Data and participants

Table 1: Participant characteristics per cohort. HYPERAKTIV controls are psychiatric patients without ADHD (clinical controls); DEPRESJON and PSYKOSE share the same 32 healthy controls. Sex, age band and recording days from the harmonised OBF-Psychiatric metadata; day-windows are the analysis units after the 80% validity threshold.

Cohort	Group	n	Female, n (%)	Age (modal band)	Recording days, median (range)	Valid day-windows	Clinical scale, median (IQR)
depresjon	cases (depression)	23	10 (43)	45-49	13 (5-18)	405	MADRS 24 (IQR 18-26)
depresjon	controls (healthy_control)	32	20 (62)	50-54	13 (8-20)	739	-
psykose	cases (schizophrenia)	22	3 (14)	35-39	13 (12-14)	390	BPRS 51 (IQR 44-58)
psykose	controls (healthy_control)	32	20 (62)	50-54	13 (8-20)	739	-
hyperaktiv	cases (adhd)	45	21 (47)	30-39	7 (2-10)	343	ASRS 49 (IQR 40-55)
hyperaktiv	controls (clinical_control)	40	20 (50)	17-29	7 (6-10)	329	ASRS 34 (IQR 25-43)
obf_psychiatric	cases (adhd, clinical, depression, schizophrenia)	130	54 (42)	30-39	7 (2-18)	1467	-
obf_psychiatric	controls (control)	32	20 (62)	50-54	13 (8-20)	739	-

DEPRESJON: 23 patients with unipolar or bipolar depression (MADRS median 24) and 32 healthy controls; PSYKOSE: 22 inpatients with schizophrenia (BPRS median 51) and the same 32 controls; HYPERAKTIV: 45 adults with ADHD and 40 clinical controls with activity recordings, read from OBF-Psychiatric. All recorded with an Actiwatch at one-minute resolution. Hashing every participant’s activity series confirmed that the PSYKOSE and DEPRESJON control recordings are identical files and that OBF-Psychiatric contains each of the 162 participants once. One calendar day with at least 80% valid minutes was the analysis unit (1,034, 1,021 and 502 days; 1,818 unique). Labels are participant-level diagnostic groups.

3.2 Predictors, models and evaluation

Thirty-one statistical and circadian features per day (Supplementary Table S2); no demographics. Majority baseline, L_2 logistic regression, random forest (500 trees), XGBoost (300 rounds, depth 3) with hyper-parameters fixed a priori; imputation and scaling inside an `imblearn` pipeline fitted on training folds only; a 1D-CNN on the raw $\log(1+x)$ series as a supplementary arm. Metrics at window and subject level (mean probability per participant): AUROC, accuracy, macro-F1, Brier, calibration slope and intercept [16], ECE, reliability curves. Ten seeds; bias-corrected and accelerated bootstrap intervals from 1,000 resamples of participants; every run writes a manifest (commit, configuration hash, seeds, versions, data checksums) and every table below is generated from it.

E1 leaky splitting. Five-fold CV under a subject-wise splitter (no participant on both sides) and a record-wise splitter (days shuffled regardless of participant), same data and seeds; the paired difference is the inflation. An automated test forbids the record-wise splitter outside this experiment.

E2 external validation. Model fitted on all of cohort A, applied frozen to cohort B, against subject-wise CV on A. Shared controls were assigned at random to exactly one cohort (16 each). DEPRESJON $\rightarrow$ HYPERAKTIV is a deliberate label-shift control.

E3 calibration. For every model in E1 and E2, plus logistic recalibration fitted on the training cohort’s out-of-fold predictions.

E4 sex strata; E5 OBF-Psychiatric. Sex-stratified E1 performance; any-psychiatric versus control and five-class arms on the harmonised 162-participant release.

3.3 Results of Part II

Table 2: E1: discrimination under subject-wise versus record-wise cross-validation. Cells give the estimate on seed-averaged out-of-fold predictions with subject-level BCa 95% bootstrap intervals; inflation is the paired record-wise minus subject-wise difference.

Cohort	Model	Window AUROC subject-wise	Window AUROC record-wise	Window inflation	Subject AUROC subject-wise	Subject AUROC record-wise	Subject inflation
depresjon	Majority class	0.465 [0.292, 0.653]	0.491 [0.459, 0.528]	0.026 [-0.147, 0.218]	0.423 [0.300, 0.551]	0.427 [0.295, 0.555]	0.001 [-0.191, 0.202]
depresjon	Logistic regression	0.779 [0.675, 0.861]	0.845 [0.781, 0.896]	0.065 [0.024, 0.122]	0.899 [0.822, 0.969]	0.965 [0.927, 0.990]	0.065 [0.011, 0.109]
depresjon	Random forest	0.770 [0.681, 0.848]	0.877 [0.823, 0.923]	0.107 [0.058, 0.168]	0.883 [0.812, 0.948]	0.977 [0.949, 1.000]	0.094 [0.041, 0.146]
depresjon	XGBoost	0.764 [0.675, 0.842]	0.878 [0.830, 0.918]	0.114 [0.060, 0.181]	0.874 [0.795, 0.947]	0.985 [0.964, 1.000]	0.111 [0.040, 0.175]
psykose	Majority class	0.437 [0.267, 0.632]	0.500 [0.470, 0.535]	0.062 [-0.113, 0.241]	0.419 [0.278, 0.557]	0.462 [0.331, 0.593]	0.043 [-0.148, 0.227]
psykose	Logistic regression	0.907 [0.820, 0.942]	0.927 [0.875, 0.955]	0.020 [0.006, 0.041]	0.974 [0.943, 1.000]	0.994 [0.983, 1.000]	0.020 [0.000, 0.043]
psykose	Random forest	0.893 [0.798, 0.934]	0.937 [0.890, 0.962]	0.044 [0.017, 0.080]	0.956 [0.913, 0.997]	0.994 [0.983, 1.000]	0.038 [0.000, 0.074]
psykose	XGBoost	0.903 [0.810, 0.941]	0.945 [0.907, 0.969]	0.042 [0.016, 0.078]	0.969 [0.935, 1.000]	0.997 [0.991, 1.000]	0.028 [0.000, 0.059]
hyperaktiv	Majority class	0.493 [0.355, 0.628]	0.483 [0.433, 0.533]	-0.010 [-0.170, 0.148]	0.497 [0.398, 0.597]	0.443 [0.349, 0.540]	-0.054 [-0.203, 0.106]
hyperaktiv	Logistic regression	0.449 [0.360, 0.535]	0.579 [0.486, 0.661]	0.130 [0.095, 0.167]	0.424 [0.328, 0.519]	0.601 [0.502, 0.695]	0.177 [0.131, 0.221]
hyperaktiv	Random forest	0.443 [0.361, 0.534]	0.612 [0.526, 0.692]	0.169 [0.135, 0.206]	0.424 [0.333, 0.518]	0.641 [0.550, 0.722]	0.217 [0.177, 0.257]
hyperaktiv	XGBoost	0.442 [0.360, 0.524]	0.593 [0.518, 0.664]	0.151 [0.116, 0.186]	0.438 [0.337, 0.530]	0.652 [0.555, 0.732]	0.214 [0.173, 0.261]

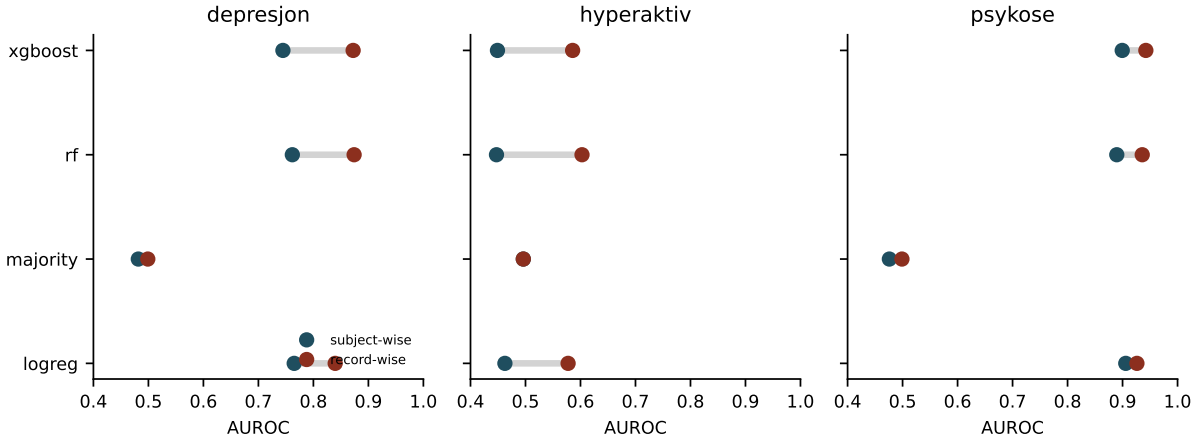


Figure 1: E1. Window-level AUROC under subject-wise (dark) and record-wise (red) cross-validation.

E1. Record-wise splitting raised window-level AUROC for every informative model in every cohort (Table 2, Figure 1). DEPRESJON: XGBoost 0.764 (95% CI 0.675 to 0.842) subject-wise versus 0.878 (0.830 to 0.918) record-wise, paired inflation 0.114 (0.060 to 0.181); subject-level leaky estimate 0.985. PSYKOSE: honest 0.893 to 0.907, inflation 0.020 to 0.044. HYPERAKTIV: honest 0.442 to 0.449 (subject level 0.424 to 0.438), record-wise 0.579 to 0.612 (subject level 0.601 to 0.652), inflation 0.130 to 0.169. Logistic regression inflated least, the ensembles most. The 1D-CNN did not exceed the feature models under honest splitting (0.696, 0.849, 0.459) and its subject-level estimate rose from 0.738 to 0.912 on DEPRESJON under record-wise splitting (Table 3).

Table 3: E1: discrimination under subject-wise versus record-wise cross-validation. Cells give the estimate on seed-averaged out-of-fold predictions with subject-level BCa 95% bootstrap intervals; inflation is the paired record-wise minus subject-wise difference.

Cohort	Model	Window AUROC subject-wise	Window AUROC record-wise	Window inflation	Subject AUROC subject-wise	Subject AUROC record-wise	Subject inflation
depresjon	1D-CNN	0.696 [0.576, 0.795]	0.729 [0.671, 0.784]	0.033 [-0.063, 0.122]	0.738 [0.634, 0.844]	0.912 [0.837, 0.959]	0.174 [0.067, 0.270]
psykose	1D-CNN	0.849 [0.744, 0.910]	0.892 [0.836, 0.935]	0.043 [-0.000, 0.095]	0.922 [0.852, 0.976]	0.986 [0.963, 1.000]	0.064 [0.016, 0.111]
hyperaktiv	1D-CNN	0.459 [0.356, 0.588]	0.513 [0.447, 0.578]	0.054 [-0.079, 0.189]	0.455 [0.371, 0.563]	0.529 [0.432, 0.622]	0.074 [-0.060, 0.208]

E2. DEPRESJON-trained models transferred to PSYKOSE without loss (external 0.769 to 0.779 versus internal 0.764 to 0.779). PSYKOSE-trained models lost 0.13 to 0.21 AUROC on

Table 4: E2: internal (subject-wise CV on the training cohort) versus external (unchanged model applied to the second cohort) performance. Window-level estimates, mean over seeds, subject-level BCa 95% intervals. EPV = minority-class subjects per candidate predictor in the training cohort.

Train → Test	Model	EPV	Internal AUROC	External AUROC	Δ	Int. slope	Ext. slope	Ext. intercept	Ext. Brier
depresjon → psykose	Majority class	0.6	0.465 [0.224, 0.662]	0.500 [0.500, 0.500]	+0.035	-4.16	0.00	-0.02	0.250
depresjon → psykose	Logistic regression	0.6	0.779 [0.701, 0.871]	0.779 [0.676, 0.847]	-0.000	0.66	0.50	0.06	0.193
depresjon → psykose	Random forest	0.6	0.770 [0.607, 0.837]	0.776 [0.688, 0.846]	+0.006	0.58	0.80	-0.02	0.193
depresjon → psykose	XGBoost	0.6	0.764 [0.608, 0.825]	0.769 [0.678, 0.835]	+0.005	0.34	0.55	-0.01	0.204
psykose → depresjon	Majority class	0.6	0.390 [0.206, 0.601]	0.500 [0.500, 0.500]	+0.110	-4.07	0.00	0.06	0.250
psykose → depresjon	Logistic regression	0.6	0.885 [0.790, 0.938]	0.689 [0.564, 0.793]	-0.196	0.72	0.27	1.20	0.262
psykose → depresjon	Random forest	0.6	0.854 [0.756, 0.926]	0.728 [0.620, 0.818]	-0.126	0.76	0.63	0.74	0.225
psykose → depresjon	XGBoost	0.6	0.877 [0.793, 0.936]	0.671 [0.569, 0.770]	-0.206	0.55	0.30	1.44	0.285
depresjon → hyperaktiv	Majority class	0.8	0.465 [0.292, 0.653]	0.500 [0.500, 0.500]	+0.035	-4.01	-0.01	0.65	0.275
depresjon → hyperaktiv	Logistic regression	0.8	0.779 [0.675, 0.861]	0.442 [0.348, 0.532]	-0.338	0.52	-0.16	-0.42	0.350
depresjon → hyperaktiv	Random forest	0.8	0.770 [0.681, 0.848]	0.461 [0.360, 0.557]	-0.310	0.69	-0.15	0.33	0.317
depresjon → hyperaktiv	XGBoost	0.8	0.764 [0.675, 0.842]	0.459 [0.358, 0.550]	-0.305	0.43	-0.09	0.48	0.364

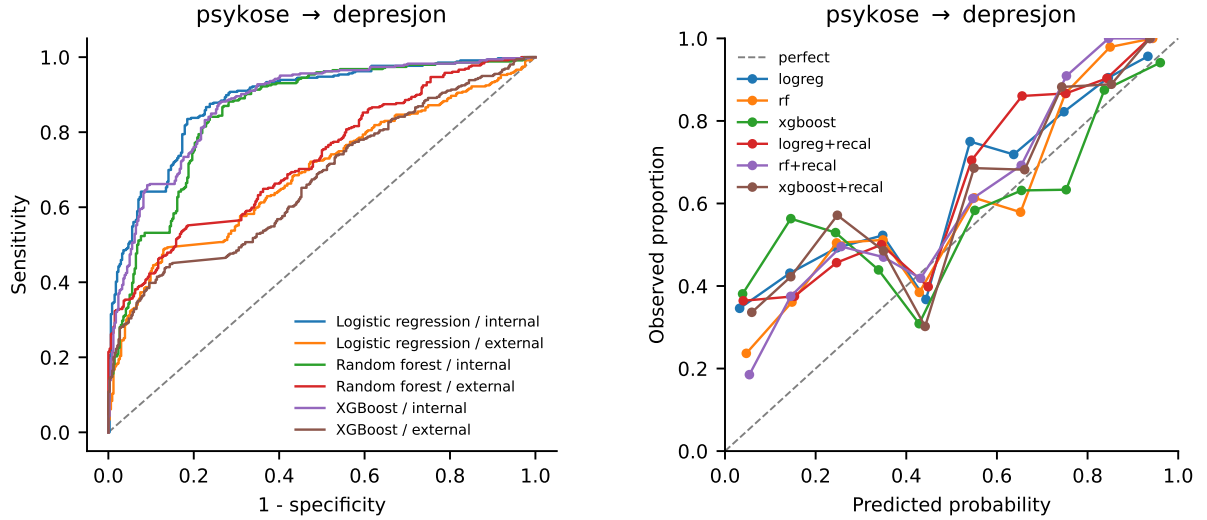


Figure 2: E2/E3, PSYKOSE → DEPRESJON: ROC (left) and reliability curves before and after recalibration (right).

118 DEPRESJON (logistic regression 0.885 $\rightarrow$ 0.689; XGBoost 0.877 $\rightarrow$ 0.671) with slopes 0.27 to
 119 0.63 and intercepts 0.74 to 1.44 (Table 4, Figure 2). The label-shift pair fell to 0.44 to 0.46.
 120 Subject-level EPV was 0.6 to 0.8.

Table 5: E3: calibration alongside discrimination for every model in E1 (subject-wise arm) and E2, plus logistic recalibration of external predictions.

Exp.	Run	AUROC	Brier	Cal. slope	Cal. intercept	ECE
E1	depresjon.logreg.subject_wise	0.779	0.187	0.69	-0.51	0.102
E1	depresjon.rf.subject_wise	0.770	0.181	0.77	0.01	0.056
E1	depresjon.xgboost.subject_wise	0.764	0.192	0.53	0.27	0.105
E1	hyperaktiv.logreg.subject_wise	0.449	0.279	-0.28	-0.01	0.161
E1	hyperaktiv.rf.subject_wise	0.443	0.281	-0.45	0.00	0.137
E1	hyperaktiv.xgboost.subject_wise	0.442	0.313	-0.23	0.00	0.221
E1	psykose.logreg.subject_wise	0.907	0.112	0.95	-0.59	0.088
E1	psykose.rf.subject_wise	0.893	0.115	1.03	-0.12	0.052
E1	psykose.xgboost.subject_wise	0.903	0.103	0.78	0.14	0.052
E2	depresjon-to-hyperaktiv.logreg.external	0.442	0.350	-0.16	-0.42	0.291
E2	depresjon-to-hyperaktiv.rf.external	0.461	0.317	-0.15	0.33	0.214
E2	depresjon-to-hyperaktiv.xgboost.external	0.459	0.363	-0.09	0.47	0.292
E2	depresjon-to-psykose.logreg.external	0.779	0.193	0.50	0.06	0.079
E2	depresjon-to-psykose.rf.external	0.776	0.193	0.81	-0.02	0.044
E2	depresjon-to-psykose.xgboost.external	0.769	0.202	0.56	-0.01	0.085
E2	depresjon.logreg.internal	0.779	0.187	0.69	-0.51	0.102
E2	depresjon.rf.internal	0.770	0.181	0.77	0.01	0.056
E2	depresjon.xgboost.internal	0.764	0.192	0.53	0.27	0.105
E2	psykose-to-depresjon.logreg.external	0.689	0.262	0.27	1.20	0.202
E2	psykose-to-depresjon.rf.external	0.728	0.225	0.63	0.74	0.149
E2	psykose-to-depresjon.xgboost.external	0.671	0.284	0.31	1.43	0.229
E2	psykose.logreg.internal	0.885	0.137	0.77	-0.10	0.061
E2	psykose.rf.internal	0.854	0.151	0.82	-0.08	0.055
E2	psykose.xgboost.internal	0.877	0.148	0.62	-0.16	0.083
E2	depresjon-to-hyperaktiv.logreg.external+recal	0.442	0.311	-0.23	0.22	0.223
E2	depresjon-to-hyperaktiv.rf.external+recal	0.461	0.301	-0.20	0.37	0.176
E2	depresjon-to-hyperaktiv.xgboost.external+recal	0.459	0.311	-0.17	0.36	0.194
E2	depresjon-to-psykose.logreg.external+recal	0.779	0.211	0.72	0.56	0.139
E2	depresjon-to-psykose.rf.external+recal	0.776	0.193	1.05	0.11	0.041
E2	depresjon-to-psykose.xgboost.external+recal	0.769	0.196	1.05	0.11	0.052
E2	psykose-to-depresjon.logreg.external+recal	0.689	0.254	0.35	0.99	0.185
E2	psykose-to-depresjon.rf.external+recal	0.728	0.221	0.77	0.67	0.129
E2	psykose-to-depresjon.xgboost.external+recal	0.671	0.258	0.49	0.93	0.200

121 **E3.** Under honest splitting, slopes were 0.53 to 0.95 and ECE 0.06 to 0.22; record-wise models
 122 looked well calibrated only because they were scoring seen participants. Recalibration on the
 123 training cohort repaired the easy transfer direction (slopes 1.05, ECE 0.04 to 0.05) but not the
 124 hard one (slopes 0.35 to 0.77, ECE 0.13 to 0.20).

125 **E4, E5.** No material sex difference where powered (Table 6; the PSYKOSE female stratum
 126 has three cases). On OBF-Psychiatric, any-psychiatric versus control reached 0.776 to 0.795
 127 subject-wise and 0.824 to 0.849 record-wise; five-class macro-AUROC 0.695 to 0.713 versus 0.770
 128 to 0.811 (Table 7).

129 4 Discussion

130 4.1 Reading Part I against Part II

131 [TODO: Once Part I is extracted: plot each study's reported accuracy against its unit of splitting;
 132 studies above the honest band with day-level or unstated splitting are the population of interest.
 133 Report how many used both DEPRESJON and PSYKOSE without recognising the shared

Table 6: E4: sex-stratified performance of the subject-wise E1 models (window level, subject-level bootstrap intervals). Subgroups with fewer than six participants or a single class are omitted.

Cohort	Model	Sex	Participants (cases)	AUROC [95\% CI]	Brier	Cal. slope	ECE
depresjon	Logistic regression	female	30 (10)	0.777 [0.649, 0.872]	0.182	0.75	0.120
depresjon	Logistic regression	male	25 (13)	0.776 [0.562, 0.882]	0.192	0.62	0.091
depresjon	Random forest	female	30 (10)	0.763 [0.623, 0.870]	0.171	0.75	0.071
depresjon	Random forest	male	25 (13)	0.777 [0.626, 0.870]	0.193	0.80	0.070
depresjon	XGBoost	female	30 (10)	0.755 [0.620, 0.859]	0.183	0.51	0.109
depresjon	XGBoost	male	25 (13)	0.772 [0.619, 0.868]	0.203	0.55	0.120
hyperaktiv	Logistic regression	female	41 (21)	0.479 [0.370, 0.602]	0.275	-0.10	0.152
hyperaktiv	Logistic regression	male	44 (24)	0.424 [0.321, 0.543]	0.283	-0.49	0.168
hyperaktiv	Random forest	female	41 (21)	0.491 [0.371, 0.611]	0.269	-0.10	0.111
hyperaktiv	Random forest	male	44 (24)	0.398 [0.278, 0.516]	0.291	-0.80	0.160
hyperaktiv	XGBoost	female	41 (21)	0.498 [0.385, 0.605]	0.293	-0.02	0.211
hyperaktiv	XGBoost	male	44 (24)	0.388 [0.278, 0.505]	0.332	-0.44	0.230
psykose	Logistic regression	female	23 (3)	0.971 [0.915, 0.998]	0.096	1.58	0.177
psykose	Logistic regression	male	31 (19)	0.890 [0.771, 0.933]	0.126	0.83	0.081
psykose	Random forest	female	23 (3)	0.966 [0.891, 0.995]	0.075	1.66	0.147
psykose	Random forest	male	31 (19)	0.874 [0.765, 0.930]	0.149	0.90	0.123
psykose	XGBoost	female	23 (3)	0.969 [0.904, 0.997]	0.059	1.16	0.100
psykose	XGBoost	male	31 (19)	0.884 [0.774, 0.933]	0.140	0.71	0.121

Table 7: E5: OBF-Psychiatric (162 participants, single control group). Transdiagnostic binary arm (any psychiatric group vs healthy control; window-level AUROC with subject-level BCa intervals) and five-class arm (macro one-vs-rest AUROC), each under subject-wise and record-wise CV.

Arm	Model	Split	AUROC / macro-AUROC	Accuracy	Macro-F1	Cal. slope
Any psychiatric vs control	Majority class	subject-wise	0.490 [0.358, 0.614]	0.641	0.391	0.25
Any psychiatric vs control	Majority class	record-wise	0.500 [0.470, 0.529]	0.641	0.391	0.25
Any psychiatric vs control	Logistic regression	subject-wise	0.795 [0.745, 0.848]	0.741	0.725	0.78
Any psychiatric vs control	Logistic regression	record-wise	0.824 [0.777, 0.867]	0.756	0.742	0.93
Any psychiatric vs control	Random forest	subject-wise	0.781 [0.726, 0.837]	0.739	0.705	0.75
Any psychiatric vs control	Random forest	record-wise	0.849 [0.811, 0.887]	0.782	0.755	1.07
Any psychiatric vs control	XGBoost	subject-wise	0.776 [0.729, 0.834]	0.734	0.702	0.63
Any psychiatric vs control	XGBoost	record-wise	0.844 [0.810, 0.885]	0.781	0.755	0.90
Five diagnostic groups	Logistic regression	subject-wise	0.713 [0.677, 0.750]	0.431	0.390	-
Five diagnostic groups	Logistic regression	record-wise	0.770 [0.740, 0.799]	0.484	0.444	-
Five diagnostic groups	Random forest	subject-wise	0.695 [0.662, 0.735]	0.434	0.350	-
Five diagnostic groups	Random forest	record-wise	0.811 [0.786, 0.843]	0.558	0.482	-
Five diagnostic groups	XGBoost	subject-wise	0.702 [0.670, 0.743]	0.448	0.355	-
Five diagnostic groups	XGBoost	record-wise	0.808 [0.785, 0.842]	0.558	0.467	-

controls.] The Part II bands are the reference: honest subject-level AUROC of 0.87 to 0.90 for depression, 0.96 to 0.97 for schizophrenia, chance for ADHD versus clinical controls; leaky bands of 0.97 to 0.99, 0.99, and 0.60 to 0.65 respectively. Reported accuracies above 0.9 for depression are reproducible, but only under the design that PROBAST+AI scores as high risk of bias.

4.2 Why the combination matters

An appraisal alone counts practices; a re-analysis alone is one more model on familiar data. Together they show, on the same cohorts, that the practices the appraisal flags are the practices that produce the reported numbers. Three findings do not depend on Part I: leakage manufactures signal where none exists (HYPERAKTIV), external validity is direction-dependent and invisible without testing (E2), and calibration is poor even when discrimination is honest and cannot be fixed from the training data alone (E3).

4.3 Limitations

One device family and country; small participant numbers with wide subject-level intervals; group labels rather than symptom trajectories; a single window length; fixed rather than tuned models (tuning inside subject-wise folds would raise honest performance somewhat without changing the paired inflation); a control group split rather than an independent one in E2; the sex analysis in PSYKOSE is underpowered; and Part I depends on what authors report.

4.4 Recommendations

For any repeated-measures mental-health dataset: state the unit of splitting; report subject-level sample sizes and metrics; report calibration slope and intercept with a reliability curve; check for shared participants across combined datasets; release code with run manifests. This repository provides a reference implementation and honest baselines for the three cohorts.

Declarations

Funding: **[TODO:]**. Conflicts of interest: none declared. Ethics: secondary analysis of public, de-identified data. Data: available from the dataset authors [1, 2, 3, 4]; checksums in the repository. Code: <https://github.com/rogerpanel/cv/tree/main/mlmh-appraisal> (**[TODO: tag, Zenodo DOI]**).

Supplementary material

S1 TRIPOD+AI checklist; S2 feature definitions; S3 search log; S4 seed study list; S5 per-window predictions and manifests.

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